

# MAGUROL®

(Doxazosin)

## PHARMACEUTICAL FORM

Tablets for oral administration available in two different strengths:

*MAGUROL® tablets 2mg/tab*: White or almost white, round, flat, tablets with diameter 7.0 mm. Each tablet contains doxazosin mesilate equivalent to 2mg doxazosin.

*MAGUROL® tablets 4mg/tab*: White or almost white, round, flat, scored tablet with a diameter of 8.5 mm. Each tablet contains doxazosin mesilate equivalent to 4mg doxazosin.

## INDICATIONS

**MAGUROL®** is indicated for the treatment of hypertension where it can be administered alone or in combination with other antihypertensive agents, such as diuretics, beta-blockers, calcium antagonists or ACE inhibitors.

**MAGUROL®** is also indicated for the treatment of urinary outflow obstruction and symptoms associated with benign prostatic hyperplasia.

## DOSAGE AND ADMINISTRATION

*Hypertension*: The daily dose ranges from 1 to 16mg. The drug is administered once a day in the morning or in the evening. The initial dose is 1mg/day. Dosage can be increased to 2mg/day after 1 or 2 weeks of therapy depending on the patient's haemodynamic condition. If necessary dose can be gradually increased to 4, 8 or even 16 mg/day, so as to achieve the best therapeutic effect. The maximum daily dosage is 16 mg.

*Benign prostatic hyperplasia*: The usual dosage range is 1 to 8mg once a day. The initial dose is 1mg/day, administered in the morning or in the evening. Depending on the individual patient's response, dosage may be then increase, gradually to 2mg after 1 to 2 weeks of treatment and if necessary up to 8mg/day, which is the maximum recommended dose. In the majority of patients therapeutic efficacy is achieved with a dose of 2 to 4 mg daily.

*Dosage in elderly patients and patients with renal failure*: No dosage adjustment is necessary

*Dosage in hepatic impairment*: The drug should be administered with caution in patients with impaired liver function, since it is known that Doxazosin is metabolized in the liver (See precautions)

*Use in children*: Doxazosin is not recommended for paediatric use.

## CONTRAINDICATIONS

Doxazosin is contraindicated in individuals with a known hypersensitivity to quinazolines. (e.g. prazosin, terazosin), or to any of the other components of the tablet.

## WARNINGS AND PRECAUTIONS

MAGUROL® should be used with caution in patients with hepatic impairment.

In patients with renal impairment and/or elderly patients no dosage adjustment is necessary.

## INTERACTIONS

Even though, doxazosin is highly bound to plasma proteins, *in vitro*, data in human plasma indicates that doxazosin has no effect on protein binding of the drugs tested (digoxin, phenytoin, warfarin or indomethacin). No adverse drug reactions have been observed with thiazide diuretics, frusemide, beta-blockers, NSAID's, antibiotics, oral hypoglycaemic drugs, uricosuric agents or anticoagulants.

## PREGNANCY AND LACTATION

Pregnancy category: C

Animal studies revealed no teratogenic effects. When the administered doses were 300 times more than the maximum recommended human dose, reduced foetal survival was observed. Since the safety of doxazosin has not been established, it should be prescribed during pregnancy only if clearly needed.

Animal tests revealed that doxazosin accumulates in breast milk. Since there is no experience in humans doxazosin is contra-indicated to nursing mothers.

## EFFECTS ON THE ABILITY TO DRIVE AND USE MACHINERY

Doxazosin may cause side effects such as dizziness, vertigo or rarely faintness thus the ability to drive or operate machinery may be impaired, especially at the beginning of the treatment.

## ADVERSE REACTIONS

**Hypertension:** In patients with hypertension receiving doxazosin the most common adverse reactions were those of postural type, rarely associated with fainting. Other adverse reactions are: dizziness, headache, fatigue/malaise, postural dizziness, vertigo, oedema, asthenia, somnolence, nausea and rhinitis. Extremely rare cases of urinary incontinence were reported and this may be related to the pharmacological action of doxazosin.

Isolated reports of priapism have been reported to be associated with  $\alpha_1$ -antagonists, including doxazosin.

Additional adverse reactions have been reported but it is not quite clear if they are due to doxazosin. These include: tachycardia, palpitations, chest pain, angina pectoris, myocardial infarction, cerebrovascular accident and cardiac arrhythmia.

**Benign prostatic hyperplasia:** Clinical trials indicate that BPH patients receiving doxazosin have the same adverse reactions as hypertensive patients.

## OVERDOSAGE

In case of overdosage symptoms that appear are mainly due to the antihypertensive activity of doxazosin. There is no specific antidote. Treatment is symptomatic and supportive. Gastric lavage or administration of activated charcoal may inhibit the absorption of the drug. To treat hypotension, liquids should be administered intravenously. Dialysis is not recommended since doxazosin is highly bound to plasma proteins.

## PHARMACOLOGY

### Pharmacodynamics

Doxazosin is a vasodilator; it exerts its effect by selective and competitive blockade of the post-junctional  $\alpha_1$ -adrenoreceptors.

The administration of Doxazosin results in a reduction of arterial blood pressure due to decrease in systemic vascular resistance. Once a day administration results in significant reduction of blood pressure, which is maintained 24 hours post-dose. At the beginning of the treatment the reduction in blood pressure is gradual and orthostatic side effects that occur are similar to those of other antihypertensive agents. The maximum reduction of blood pressure occurs 2 to 6 hours after administration and it is the same in both the supine and the standing position.

In contrast with the non-selective  $\alpha$ -adrenoreceptor blocking agents, no tolerance is developed after long-term administration of doxazosin. Tachycardia is hardly seen after long term treatment with the drug.

A modest reduction in plasma total cholesterol, LDL-cholesterol and triglyceride concentration was observed after long-term treatment with doxazosin which is of benefit to hypertensive patients with concomitant hyperlipidaemia.

Doxazosin has no adverse metabolic reactions which make it suitable for patients with asthma, diabetes mellitus, insulin resistance or gout.

MAGUROL® uptake by patients with benign prostatic hyperplasia results in a significant improvement in urodynamics and both obstructive and irritating symptoms. This effect results from selective blockade of the  $\alpha$ -adrenoreceptors located in the prostatic muscular stroma, capsule and bladder neck, thus the contraction of smooth muscle is inhibited and the resistance in urinary flow is decreased.

### Pharmacokinetics

After oral administration doxazosin is well absorbed from the gastrointestinal tract, and peak plasma levels occur after 2 to 3 hours. The bioavailability of the drug is 65% which indicates first pass metabolism of doxazosin by the liver. Concomitant administration with food reduces slightly its absorption but this has no clinical significance.

Doxazosin is highly bound to plasma proteins (about 98%)

It is metabolised in the liver mainly via O-demethylation of the quinazoline nucleus or by hydroxylation of the benzodioxan moiety. Approximately 63% of the administered dose is excreted in faeces and 9% in urine. Only 5% of the administered dose is excreted as unchanged drug in the faeces. The remaining percentage is excreted in the form of metabolites.

The elimination of doxazosin is biphasic. The elimination half-life of the drug is 22 hours.

The pharmacokinetic behaviour of doxazosin in elderly patients or in those with renal impairment is the same as in young normal volunteers. Doxazosin should be given with particular caution to patients with hepatic impairment, since it is almost exclusively metabolised and excreted by the liver.

## PHARMACEUTICAL PARTICULARS

**List of the excipients:** MAGUROL® tablets contain as excipients: Lactose, cellulose microcrystalline, sodium starch glycollate, magnesium stearate, sodium lauryl sulphate.

**Incompatibilities:** None known

**Shelf-life:** 24 months

**Special precautions for the storage of the product:** Store below 30°C in the original package.

**Packaging:**

MAGUROL® tablets 2 mg/tab: Box of 20 tablets (2 blisters PVC-PVDC/Al of 10 tablets) and 1000 tablets (100 blisters PVC-PVDC/Al of 10 tablets)

MAGUROL® tablets 4 mg/tab: Box of 20 tablets (2 blisters PVC-PVDC/Al of 10 tablets) and 1000 tablets (100 blisters PVC-PVDC/Al of 10 tablets)

**Instructions for the patient:** None

**NAME AND ADDRESS OF THE MANUFACTURER OF MAGUROL 2MG TABLETS:**

Medochemie Ltd, (Central Factory), 1 – 10, Constantinoupoleos Street, 3011 Limassol, Cyprus

**NAME AND ADDRESS OF MANUFACTURER OF MAGUROL 4MG TABLETS:**

Medochemie (Far East) Ltd. (Oral Facility), No. 40 Street 6 Vietnam Singapore Industrial Park II, Binh Duong Industry Service Urban Complex, Hoa Phu ward, Thu Dau Mot city, Vietnam

**NAME AND ADDRESS OF PRODUCT REGISTRATION HOLDER IN MALAYSIA**

Komedic Sdn Bhd, 4, Jalan PJS 11/14, Bandar Sunway, 46150 Petaling Jaya, Selangor – Malaysia

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